Machine Learning
Project Specifications
CEMCLRN / MAC-LRN
1T2627
Dr. Ann Franchesca Laguna

End-of-Term Project: Multi-Tier Machine Learning System

1. Objective

The goal of this project is to **solve a real-world problem** by designing, implementing, and documenting a complete, end-to-end machine learning system that spans at least two distinct computing environments. This project is a capstone experience that will allow you to apply the principles of data engineering, model deployment, and system architecture covered in this course. You will be evaluated not just on the performance of your machine learning model but on the robustness of your entire system.

2. Project Requirements

Your final system must integrate at least two of the following environments: **Cloud**, **Edge**, **Mobile**, or **TinyML**.

Your project must fulfill the following technical requirements:

- **Problem Statement:** Clearly define the problem you are solving. This should be a tangible, real-world issue that can be addressed with a machine learning solution.
    - *Examples:* "Real-time human presence detection in a room to optimize energy usage," or "Automated identification of different types of garbage for a smart recycling bin."
- **System Tier 1 (Data Collection/Local Inference):** Design and implement a component that operates on a resource-constrained device (Edge, Mobile, or TinyML). This component must perform one of the following tasks:
    - Continuously collect data (e.g., sensor readings, images) relevant to your problem.
    - Run a lightweight, pre-trained model for real-time inference (e.g., simple classification, anomaly detection).
- **System Tier 2 (Cloud or Edge Processing/Training):** Design and implement a component that runs on a cloud-based or Edge-Based server. This component must:
    - Ingest data from your first tier.
    - Train a more complex machine learning model based on the collected data.
    - Store and manage the trained model for deployment.
- **Version Control:** You must use Git for all code and non-binary file version control. Your repository must be publicly accessible (e.g., on GitHub, GitLab) and include a well-structured commit history with meaningful commit messages. Use a `.gitignore` file to exclude large datasets and other unnecessary files.
    - **Team Contribution:** If working in a team, each member must commit their individual contributions. If a contribution is done by multiple members, it should be explicitly stated in the commit message (e.g., "Co-authored-by: [Member 1 Name] and [Member 2 Name]"). Each team member should commit at least once.
- **Project Deliverables:**
    - **Git Repository:** A link to your public repository containing all project code and configuration files.

- **Project Report (PDF):** A 5-7 page report using the ACM conference format detailing your problem statement, system design, implementation choices, challenges, and results. This report should include:
  - An architectural diagram of your system.
  - A clear description of your chosen environments and frameworks (e.g., TensorFlow Lite on the Edge, PyTorch on the Cloud).
  - Justification for your design decisions (e.g., "We chose this architecture to minimize data transfer costs").
  - A summary of your model's performance metrics (e.g., accuracy, latency).
- **Final Demonstration:** A live presentation (15 minutes) that showcases your end-to-end system in action, demonstrating the data flow and model functionality across the two tiers.

3. Important Notes

- The focus of this project is on **systems integration**, not on achieving state-of-the-art model performance. A simple classification or regression model is perfectly acceptable as long as it is integrated into a robust, multi-tier system that solves your problem.
- You can borrow hardware from the CTLab; tell the technician that you are part of the class. If you need to take a device home, please inform the instructor first.
- Feel free to leverage existing datasets and pre-trained models; however, the core of your work must be the design and implementation of a complete, end-to-end system.
- The grade of a team member who is not contributing can be less than the others up to a grade of zero.
- Any group can drop their group member if they are completely not contributing.